

A Study of Pulsed-Charging Methods for Floating Gate MOSFETs Designed for Radiation Measurements

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Abstract—In this paper, two pulse-based injection methods are employed for charging/discharging a floating gate transistor, specially designed to work as dosimeter in radiotherapy applications. When constant voltage pulses are applied to the structure, the floating gate can be successfully charged, but the injected charge self-limits the injection current, slowing down the injection process. To overcome that, an adaptive voltage pulse technique is considered, showing a roughly constant injection current and good enough control of the charge in the floating gate.

Index Terms—Floating-Gate transistor; Fowler-Nordheim; Dosimetry; Pulse generator

I. INTRODUCTION

The International Commission for Radio-Protection (ICRP) recommends in-vivo dosimetry for safety and quality assurance in radiotherapy applications [1], [2]. In-vivo dosimetry is an important practice to prevent accidents and also to ensure the effective execution of the planned treatment. In this context, the development of highly accurate dosimeters is mandatory. Recommendations suggest that the delivered dose should not vary from the planned treatment more than 3% in each session. In this context, it is necessary for the dosimeters to be highly accurate.

For the last decades, MOS dosimeters have been proposed and used for medical applications as they present many advantages [3], [4]. The main effect of radiation is the shift of the threshold voltage (V_T). Measuring the change in V_T , the absorbed dose can be estimated and the MOSFET can be used as a dosimeter [5]. Nevertheless, there are still a series of issues that must be addressed. In the last years, we have proposed the use of Field Oxide as a thick gate oxide (FOXFET) to overcome some of these issues, proving that the dosimeter sensitivity can be improved [6]. The FOXFET has shown to be appropriate for radiotherapy [7], even though the thick gate oxide exhibits instabilities in the trapped charge, known as fading. The FOXFET was also used to implement

differential measuring methods [8]–[10], which implies the use of different gate voltages applied to each transistor. For that, an electric connection with the dosimeter is needed during irradiation, which is undesired in medical applications.

The floating gate (FG) MOSFET is an alternative device to be used as a dosimeter [11]–[15]. When it is exposed to radiation, electron-hole pairs are created in the oxide and the generated charge is collected by the FG, altering the drain current value as a consequence. By measuring the current variation, before and after exposure, the device's sensitivity to radiation can be obtained. The FGMOSFET sensitivity can be improved extending the floating gate over the field oxide [12], [13], in a similar way as the FOXFET, but without the instabilities, as the active device is a standard transistor with a thin gate oxide. Moreover, it also presents the advantage that if the FG is charged, the electric field within the oxide can be maintained without external power supply [13], getting rid of the electrical connection during irradiation.

The charge in the floating gate defines the sensitivity to radiation of the device. The more the charge, the higher the sensitivity and also the dose range of the dosimeter [16]. Moreover, if the device is discharged, it can be charged again in order to be re-used. Hence, the charging process of the floating gate is very important in dosimetry applications [17]. The charging of floating gate devices is also very common in memory applications. Here the device exhibits a region with a thinner oxide, called the tunnel oxide, where a high voltage pulse is applied and a tunnel current charges the FG. In 1-bit memories, the exact amount of charge injected is not important as long as two states can be differentiated. In multi-level memories, more charge states should be distinguished and the charging process should be more controllable. The Incremental Step-Pulse Programming (ISPP), a process where the voltage in the injection pulse is increased gradually, is proposed not only for a better differentiation of charge, but also to improve programming times, and to overcome some difficulties in die-to-die and cell-to-cell variations in the charging characteristics

of the FG memory cells [18].

In this paper, we study the possibility of applying a pulse charging method to a Floating Gate transistor designed for dosimetry applications. Two methods are evaluated and compared, a fixed voltage pulse and an adaptive voltage pulse, similar to the ISPP method used in memories. The specific aim of this study is to determine whether the desired charge level can be achieved with either method and also to prove the time benefits of the adaptive voltage method.

II. FLOATING GATE DESIGN FOR DOSIMETRY

The Floating Gate dosimeter was designed and fabricated in a CMOS 0.5 μm process through MOSIS University Support Program. A schematic diagram of the device is shown in Fig. 1. The device is a p-channel MOSFET, where the polysilicon layer (P1L) that acts as the floating gate is extended over the field oxide to improve radiation sensitivity [6], [12], [13]. The charge induced by radiation is generated in both the gate oxide (GOX) and the field oxide (FOX) [5]. A fraction of this charge will be collected in the floating gate (Q_{FG}), giving place to a shift in the floating gate potential (V_{FG}). To improve the sensitivity of V_{FG} with radiation dose, the P1L over FOX should be much larger than the P1L over GOX, hence a ratio $A_{FOX}/A_{GOX} = 100$ was chosen. A sensitivity of 6.5 $\mu\text{A}/\text{Gy}$, at $I_D = 288\mu\text{A}$, was measured with gamma radiation [16].

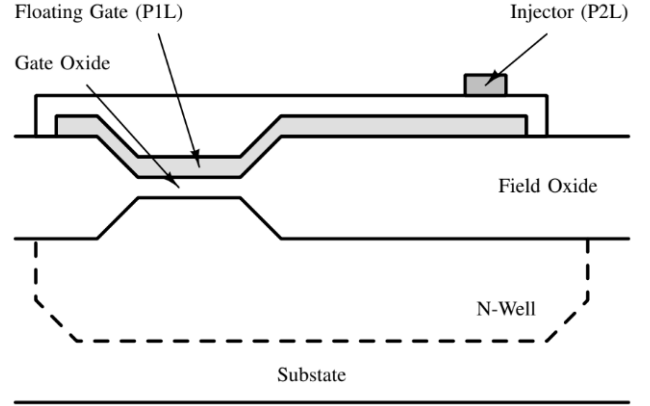
The radiation induced charge that reaches the FG will neutralize part of the charge that already existed in the FG. This means that the device should be charged before exposure to radiation. Charging the floating gate occurs by Fowler-Nordheim (FN) tunneling from the injection electrode to the floating gate [19]. When a voltage is applied to the injector (V_{inj}), the floating gate potential (V_{FG}) due to the capacitive coupling of the conducting polysilicon strips is

$$V_{FG} = \frac{C_{inj} \cdot V_{inj} + (C_{fox} + C_{gox}) \cdot V_{FET} + Q_{FG}}{C_{inj} + C_{fox} + C_{gox}}, \quad (1)$$

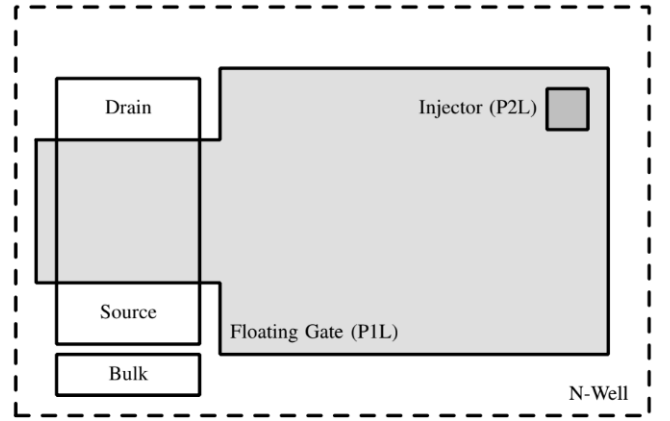
where C_{inj} is the capacitance between the injector and the floating gate, C_{fox} and C_{gox} are the capacitance of the field oxide and gate oxide respectively, V_{FET} is the voltage commonly applied to the drain, source and bulk of the FG-MOSFET, and Q_{FG} is the charge accumulated in the floating gate. For injection to occur, V_{FG} must be different to V_{inj} , hence the voltage applied to the injector should drop mainly across the interpoly oxide and not in the oxide below the FG. Considering $Q_{FG} \simeq 0$ in Eq. 1, this happens when $(C_{fox} + C_{gox}) \gg C_{inj}$.

The injection electrode was implemented with a second polysilicon layer (P2L) available, which lies above the P1L. This is a different approach compared to FG memory cells, where the charging of the FG occurs from the substrate through the tunnel oxide. The P2L area was designed as little as possible, resulting in $A_{inj} = 6\mu\text{m} \times 6\mu\text{m}$. No Control Gate was included over the Floating Gate as the parasitic capacitance of the control gate impoverishes radiation

sensitivity [12]. For more details the reader can consult [16] and [17].



(a) Floating Gate cross section view



(b) Floating Gate top view

Fig. 1: Simplified diagram for the Floating Gate physical design. (a) The cross-section shows how the different polysilicon layers overlap, and gives an idea of the different oxides involved in the structure of the device. (b) A simplified version of the layout to understand the different areas involved. For better understanding, metal, contacts and n/p diffusion layers were omitted. Bulk and Source are shorted from fabrication.

III. EXPERIMENTAL SETUP

The floating gate device was fabricated. Since the fabricated die studied was not wire-bonded nor encapsulated, the electrical characterization was performed in a probe station.

The experimental setup switched between two states: measuring drain current at a fixed source-drain voltage (V_{SD}) of 4.5V, and charge injection in the FG by sending pulses to the gate with all the other terminals shorted to a fixed voltage (V_P).

A Keithley 617 electrometer was used, both as a voltage source and as an ammeter, alongside with a Hewlett Packard HP8112A programmable pulse generator and an oscilloscope. Measurements were performed at room temperature.

When measuring current, the experimental setup used was as shown in Fig. 2a and the fixed voltage V_{SD} was set to

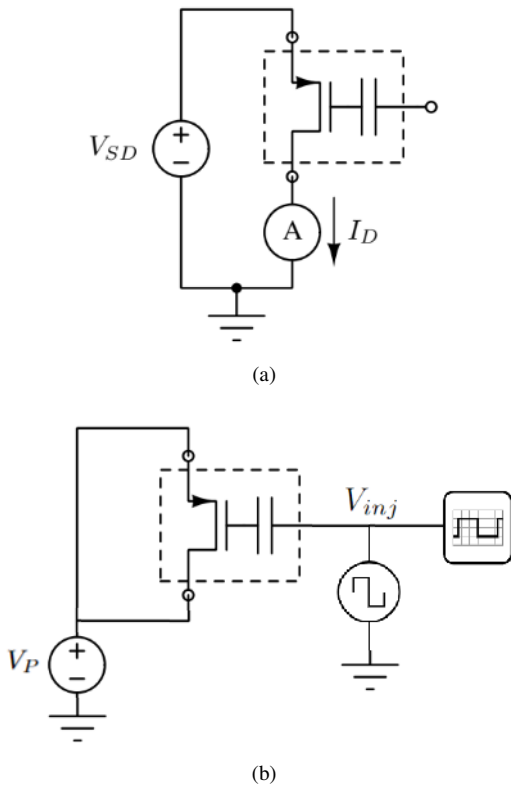


Fig. 2: Schematic circuits of the experimental setup. (a) Schematic circuit for the drain current measurement. (b) Schematic circuit for sending pulses to the injector electrode.

4.5 V. For charge injection, the setup was switched to apply the pulses as shown in Fig. 2b and V_P was set to the quiescent level of the pulse. The same voltage source was used to set V_{SD} and V_P .

A key aspect of the FGMOSFET is that the FG itself can be charged or discharged in order to change its operation state. For charge injection, the pulse generator was connected to the injection electrode. Because the Floating Gate transistor used to study the charging process was a p-channel device, it had to be charged with electrons to induce a conduction channel underneath the interface. Therefore, in order to charge the device, the applied pulse had to be negative with respect to V_P , whereas to discharge it, the pulse had to be positive.

Two different methods were studied. The first one involved using constant voltage pulses, while the second one involved using variable voltage pulses. The pulse width was constant and the same for both methods, which will be described in detail in the following section.

IV. MEASUREMENTS

In order to start charging the FG, a pulse width and height had to be determined, considering that injection occurs due to the Fowler-Nordheim (FN) tunneling mechanism, which sets a minimum voltage that must be overcome in order for injection to happen. FN tunneling can be modeled according to

the expression in Eq. 2 for the injection current as a function of the electric field applied to the oxide [19], [20]:

$$I_{inj}(V_{ox}) = A_{inj} \cdot C \cdot E_{ox}^2 \cdot \exp\left(-\frac{E_0}{E_{ox}}\right) \quad (2)$$

where C and E_0 are constants in terms of the effective mass and barrier height [19], $E_{ox} = \frac{V_{ox}}{t_{ox}}$ is the electric field across the oxide, $V_{ox} = V_{inj} - V_{FG}$ is the voltage across the interpoly oxide, V_{inj} is the voltage of the injection electrode, V_{FG} is the voltage of the floating gate, and t_{ox} is the interpoly oxide thickness.

From the injection current dependance to the voltage across the injection electrode a minimum voltage of around 25 V was estimated to be the value to overcome in order for injection to happen [17].

A. Charging with constant voltage pulses

Before starting the first test, an initial current of 11 nA was measured. The characteristics of the pulse to be applied to the injector electrode were chosen to be: *pulse width* $t_{ON} = 300$ ms; *pulse height* $V_{pulse} = V_{inj} - V_P = -26.1$ V. The pulse width was chosen in order to produce a $\Delta V_{FG} \simeq 100$ mV after each pulse for the uncharged device.

After applying the first pulse, the current went up to $0.85 \mu\text{A}$, and after the second pulse, it went up to $1.61 \mu\text{A}$, as it can be seen in Fig. 3a and Fig. 3b, where the current measured and the height of the pulses is shown, respectively. As more pulses were applied, the current increase rate decreased, until the current variation was negligible. This is explained considering that while the floating gate is being charged, V_{FG} changes and therefore V_{ox} changes as well, and because the Fowler-Nordheim current through the interpoly oxide is a function of the voltage applied to the structure, what this ultimately means is that if the pulse height is maintained, V_{ox} is reduced, the FN current will be smaller each time and the injection level will decrease.

Fig. 6 shows that after applying 213 pulses, the drain current reached $I_D = 33 \mu\text{A}$.

Although this procedure works, the charging becomes slower as more pulses are needed to take the Floating Gate to a desired operation state. Also, because the amount of injection current is constantly changing, it is harder to predict how many pulses will be required to reach a desired drain current. Moreover, the injection process can saturate if V_{FG} increases so that E_{ox} is not enough for FN current to flow. Hence, we decided to try a different technique: varying the height of the pulses in order to maintain V_{ox} practically constant.

B. Charging with variable voltage pulses

In view of the fact that the first method showed a decline of the injection current as more pulses were applied at the injector electrode, the goal of the second method was to maintain that current at a constant value. By doing this, we would expect that the floating gate potential changed with a constant slope, needing a lower number of pulses to charge the FG compared

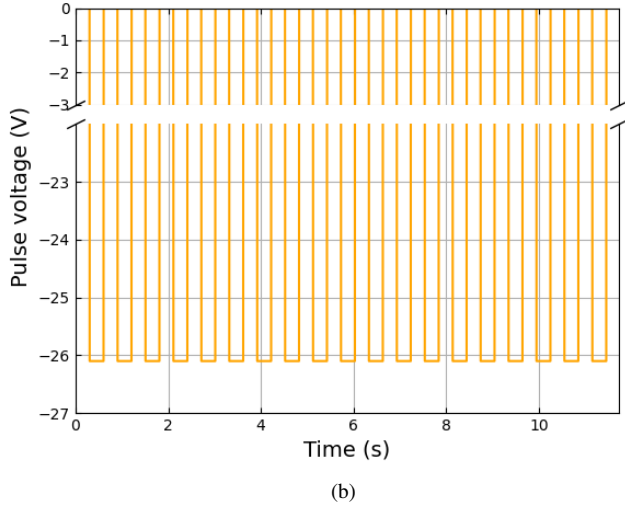
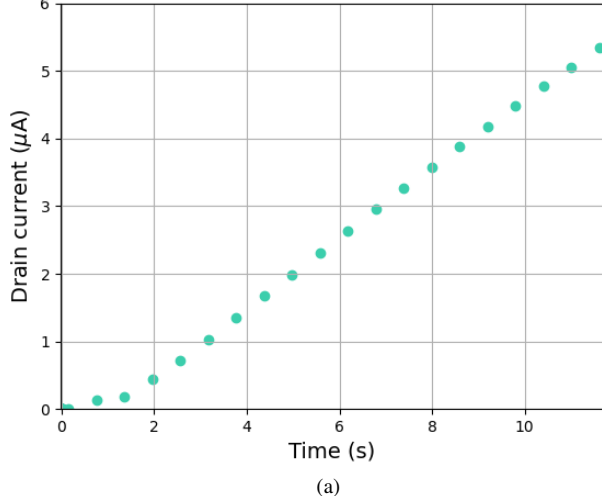


Fig. 3: Measurements of the first method. (a) Drain current variation. (b) Constant pulses applied to the injector electrode.

to the previous case, and that a higher drain current could be reached.

To relate the measured drain current to the floating gate's voltage V_{FG} , the transfer characteristic of a p-channel MOSFET, illustrated in Fig. 4, with identical dimensions to the FGMOSFET studied was used as reference.

The floating gate device tested was the same as the one studied with the previous method. Before testing this second method, the FG was discharged, applying positive pulses, until the measured drain current was again 11 nA. The characteristics of the first pulse were the same as before: *pulse width* $t_{ON} = 300$ ms; *pulse height* $V_{pulse} = -26.1$ V.

The amount of charge injected (ΔQ) and the average injection current ($\bar{I}_{inj}$) can be estimated using the expressions in Eq. 3 and Eq. 4, respectively:

$$\Delta Q = (C_{gox} + C_{fox}) \times \Delta V_{GS} \quad (3)$$

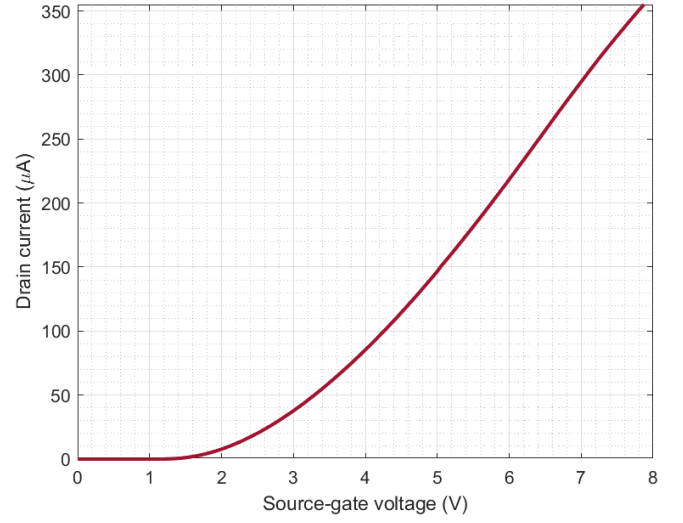


Fig. 4: Transfer characteristic curve of a standard PMOS transistor for $V_{SD} = 4.5$ V, used as reference.

$$\bar{I}_{inj} = \frac{\Delta Q}{t_{inj}} \quad (4)$$

It is clear that the average injection current will be approximately constant if the gate to source voltage variation (ΔV_{GS}) and the time duration of the injection (t_{inj}) stay the same.

Keeping that in mind, and considering that the FN injection current depends on the voltage across the oxide V_{ox} , only the voltage height V_{pulse} of the following pulses was updated. Since the goal was to keep V_{ox} constant, the new value of the pulse height would have to be changed compensating the variation of the floating gate voltage V_{FG} before and after the previous pulse was applied. By doing this, we would expect the FN current and the injection level to remain unchanged.

After measuring the initial current and applying the first pulse, the voltage increase of the following pulses were calculated as will be described next.

For each drain current value measured, a corresponding voltage V_{GS} was estimated observing the transfer curve shown in Fig. 4 giving as a result a voltage variation ΔV_{GS} , which would be the voltage increase of the following pulse. Then, if V_{pulse_N} was the voltage height of the N^{th} pulse, then the voltage height for the upcoming pulse, $V_{pulse_{N+1}}$, would be:

$$V_{pulse_{N+1}} = V_{pulse_N} + \Delta V_{GS_N}$$

with:

$$\Delta V_{GS_N} = V_{GS}(I_{D_{N+1}}) - V_{GS}(I_{D_N})$$

where I_{D_N} is the drain current measured before applying the N^{th} pulse and $I_{D_{N+1}}$ the drain current measured before applying the $(N+1)^{th}$ pulse.

The results obtained are presented in Fig. 5a and Fig. 5b, where the current measured and the height of the pulses is shown, respectively. In this case, $\Delta V_{GS} \simeq 100$ mV for all

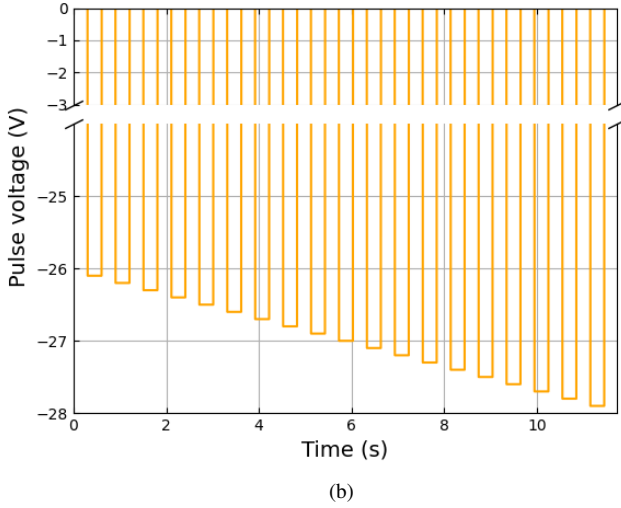
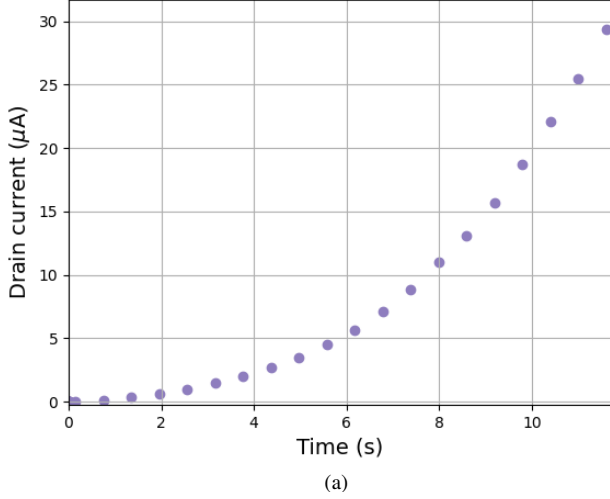


Fig. 5: Measurements of the second method. (a) Drain current variation. (b) Variable height of the pulses applied to the injector electrode.

pulses. The measurements showed that the number of pulses needed to charge the FG to a level of current close to the $33\mu\text{A}$ seen before decreased to only 20. This demonstrates that the device behaved as expected and that the second method is a more complex, yet better, approach to charge the FG. Moreover, after 52 pulses the drain current reached $I_D = 248\mu\text{A}$, which would not be possible using the constant pulse method.

Fig. 6 shows a comparison of the measured drain current, for each method studied, against the number of pulses applied to the injector electrode.

V. DISCUSSION AND CONCLUSION

This study described a floating gate transistor designed for dosimetry applications and presented two charging methods using a pulse generator. The main difference between the two

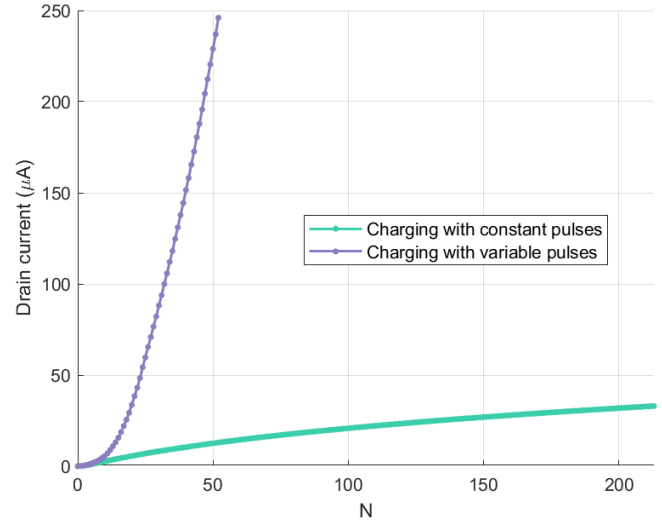


Fig. 6: Comparison between the two methods: drain current against number of pulses applied to the injector electrode.

methods was that one involved applying constant pulses to the injector electrode while the other involved applying pulses of variable voltage height. The main result was that it is possible to charge the device employing pulses, regardless of the method.

Both methods were manually tested, as these were the first trials. Yet, the implementation of the charging and discharging of the floating gate in an automated way is already being studied.

The first method tested was found to be simpler, because the pulse characteristics did not change, which would make the development of an automated process easier since the pulse generator's settings would need to be set only at the beginning. However, the amount of pulses, and consequently the time, needed to charge the device was considerable. Another limitation of this method is due to the fact that as the floating gate is charged the voltage across the oxide decreases and, with it, the Fowler-Nordheim injection current, eventually leading to a point where the floating gate is no longer being charged. For instance, it would be practically impossible to charge the device from 10 nA to $200\mu\text{A}$ employing this method. The results would be different if the pulse characteristics were modified, such as the time width, but the limitation would still exist.

The second method tested, which was more complex compared to the first one, proved to be better if time restrictions are considered. Because the injection current remains constant, this technique allows to reach higher drain currents faster and in a more controlled way. Nonetheless, a restriction found in this method was that it depends on the pulse generator's high and low level limits, i.e the maximum pulse height. This needs to be considered alongside with the physical characteristics of the device, as the Fowler-Nordheim tunneling effect depends on parameters such as the interpoly oxide thickness, impacting on how high the height of the pulse should be. Finally, it

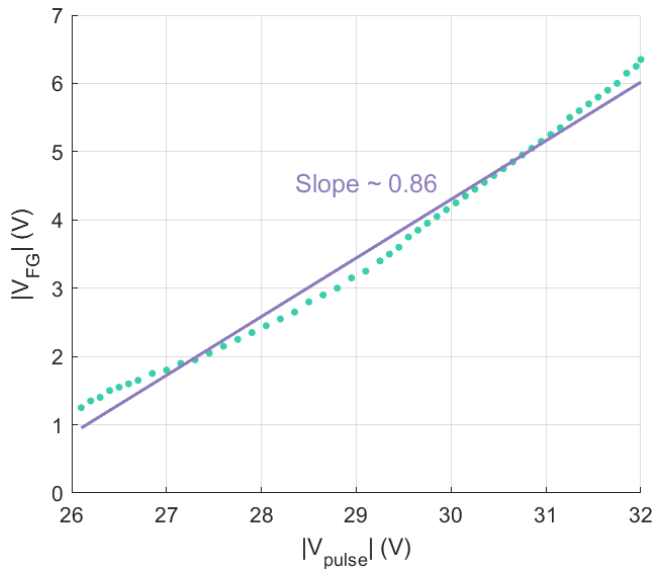


Fig. 7: Change in the Floating Gate potential with applied pulse height in the variable voltage method. The linear fit shows the calculated slope.

should be mentioned that the change in the pulse height was expected to match the change in the FG voltage, showing a slope = 1. This was not the case as shown in Fig. 7, and the slope was approximately = 0.86 showing degradation. This effect is already known in memory applications with ISPP methods [21], [22] and deserves a deeper study.

Overall, even though the injection method must be improved, results shown in this communication suggest that a precise control of the FG charge can be achieved when a variable voltage pulse technique is used. In the future, the pulse height variation does not need to be constant, and it can be determined in order to achieve a desired amount of charge. Moreover, a combination of positive and negative pulses can be used for more precision. Another important aspect to evaluate would be the time width that the pulses should have. In this study, 300 ms were considered because it was empirically seen that it was enough for the tunneling effect to occur. An even more controlled and precise approach than the ones presented in this paper might include varying not only the height of the pulses but also its time width, allowing for less error when desiring to make the floating gate get to a certain operation point.

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